

CHAPTER – II

IDEATION PHASE

1.2 – Problem Statement

Agriculture is the backbone of India's economy. It provides jobs for 58 percent of the workforce. Agriculture also contributes a lot to India's GDP and export earnings. India is a producer of many crops. These include rice, wheat, sugarcane, cotton and pulses. Crops are grown across three seasons. Kharif, Rabi and Zaid. They are grown in 36 states and union territories. Each place has weather conditions.

The agriculture sector has a problem. There is no system to track crop production data. This data is needed at the state and district levels. There is a lot of data on crop production. Many government agencies have this data. These include the Ministry of Agriculture and Cooperation NABARD and the India Meteorological Department. However, this data is not organized. Each agency uses names for columns. They also use codes for districts and states. Analysts and policymakers use Excel to analyse data. This is time-consuming and prone to errors. They cannot update data in time.

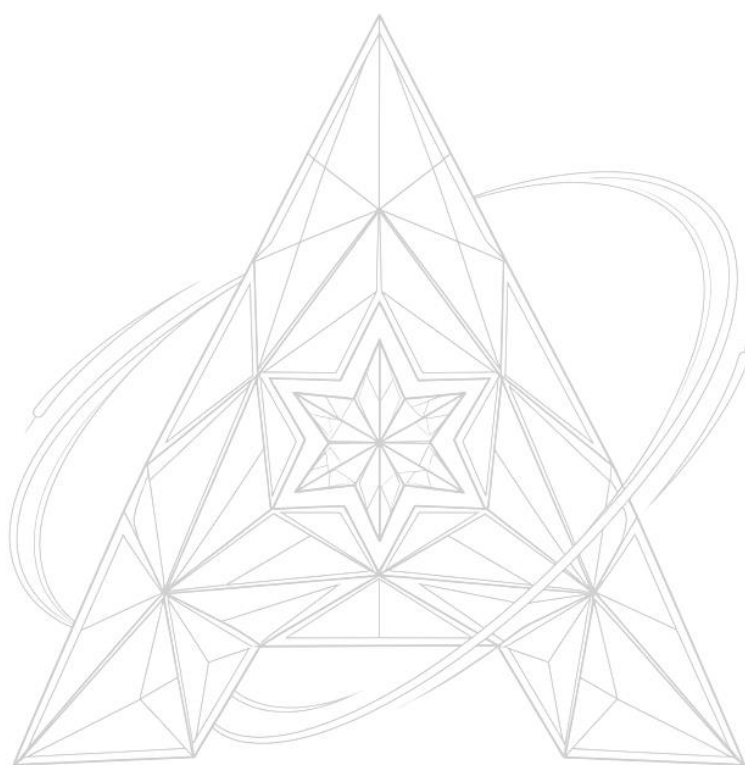
The lack of analysis tools has serious consequences. The central government cannot quickly identify states with crop yields. State agriculture departments lack tools to target subsidies and seeds efficiently. Farmer cooperatives cannot compare production trends. The main problem is not a lack of data. It is the lack of a platform to bring data together and make it visual. Tableau can solve this problem. It is a business intelligence tool. It can combine data from sources. It can also make data visual and interactive.

This project aims to develop a Tableau-based system. It will consolidate data from government sources. The data will cover 1997 to 2024. It will include data on crops. The project will build dashboards. These dashboards will allow users to filter data by crop, season, state, district and year.

The expected outcomes are both practical and strategic. The project will deliver a data pipeline and dashboards. These will reduce analysis time. Eliminate manual reporting. The

project will also make trends and patterns visible. This will enable evidence-based decisions, on procurement, irrigation and agricultural investment.

In summary Indias agricultural data is rich but hard to access. Better analytical tools are needed. Tableau can solve this problem. This study shows how.



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